PEARSON EDEXCEL INTERNATIONAL GCSE (9–1) · PHYSICS · 4PH1/2P

A* Examination Guide

Built from 17 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the Paper 2P question papers and mark schemes. Model answers follow the marking points examiners list, and every warning comes from an “allow”, “ignore”, “reject”, “condone” or “−1 for POT error” note in the mark schemes. Paper 2 examines the whole specification including the bold, Paper-2-only content: momentum, moments, electrostatics, transformers, specific heat capacity, gas laws in kelvin, redshift and the Big Bang, and the HR diagram.

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4 Momentum
5 Moments
6 Electrostatics and electromagnetism
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12 Practical methods and final checklist

Papers analysed: Jan 2019 (legacy 4PH0), Jun 2019, Jan 2020, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 HOW PAPER 2P IS BUILT

Format: 1 hour 15 minutes, 70 marks (the legacy 4PH0 paper was 1 hour, 60 marks), 7–9 questions: just over 1 minute per mark. Each paper has one long practical method (5–6 marks), two or three multi-step calculations, and several 2–4 mark explanations. Paper-2-only equations (momentum, moments, transformers, specific heat capacity, gas laws, redshift) are central.

Topic	Papers (of 17)	What is asked most
Moments and centre of gravity	13	moment = Fd, principle of moments, forces on supports
Momentum and collisions	12	p = mv, conservation, F = Δp/t, safety features
Electrostatics and charge	13	charging by friction (electrons), earthing, Q = It, E = QV
Specific heat capacity and thermal energy	11	ΔQ = mcΔT, the SHC experiment, heating curves
Particles, states and gas laws	11	solid/liquid/gas, melting plateau, p/T in kelvin
Astrophysics and cosmology	15	HR diagram, absolute magnitude, redshift, CMBR, Big Bang
Energy resources and stores	12	advantages/disadvantages, store-to-store transfers in power stations
Sound and oscilloscope	10	speed of sound methods, period from a trace, f = 1/T, hearing range
Transformers and electromagnets	8	turns ratio, VpIp = VsIs, how a transformer works, transmission
Nuclear fission and fusion	10	chain reaction, fusion conditions, products

2 TWENTY MARK-SCHEME RULES

1. **Calculations:** substitution, rearrangement, evaluation (and a unit mark where asked). Correct answers with no working usually score full marks, but working protects you through error carried forward.
2. **POT errors cost one mark:** convert ms → s, cm → m, g → kg, kPa → Pa, hours and days → s.
3. **Momentum is p,** not m ("reject use of m for momentum"); unit kg m/s.
4. **"Momentum is conserved" alone is insufficient** for a collision explanation; say total momentum before = after.
5. **Safety features:** increase the time of impact, so the rate of change of momentum is smaller, so the force is smaller. "Absorbs the impact" is ignored.
6. **Principle of moments:** (sum of) clockwise moments = (sum of) anticlockwise moments, about the same point.
7. **Charging by friction:** electrons are transferred; never positive charges or "positive electrons". Losing electrons makes an object positive.
8. **Q and I symbols:** "C" or "c" for charge or current is rejected.
9. **Temperatures in kelvin** for gas laws (K = °C + 273); using °C scores at most partial credit.
10. **Particles:** use "particles" with speed or kinetic energy; reject "particles expand"; bonds weaken when melting.
11. **Heating curve:** temperature constant during melting (plateau) at the melting point.
12. **Absolute magnitude:** brightness of a star at a standard distance (10 parsecs); "brightness" alone is 1 mark.
13. **Big Bang evidence:** CMBR and red-shift of galaxies ("condone one missing word").
14. **Energy stores:** name stores (kinetic, gravitational, thermal, chemical, nuclear) and transfers (mechanically, electrically, by heating, by radiation); "mechanical energy" is rejected.
15. **Methods:** apparatus named (balance not "scale"), measurements, control variables, repeats and means, and the equation used.
16. **Means:** exclude anomalies and round to the precision of the data (201 W, not 200.6 W).
17. **Resources:** "cheap", "reliable" or "greenhouse gases" unqualified are ignored; name the pollutant and its effect.
18. **Hearing range:** 20 Hz to 20 000 Hz with consistent units.
19. **Transformers only work with a.c.;** say the alternating current produces a changing field.
20. **Data conclusions:** calculate the constant (e.g. $B \times d^3$) for at least two rows and compare before concluding.

3 EQUATIONS FOR PAPER 2

Quantity	Equation	Quantity	Equation
momentum	$p = m \times v$	force (change of momentum)	$F = (mv - mu) \div t$ (given)
moment	moment = $F \times d$	charge	$Q = I \times t$
energy (charge)	$E = Q \times V$	power	$P = W \div t$
transformer	$Vp/Vs = Np/Ns$	power in = out	$Vp\ Ip = Vs\ Is$
thermal energy	$\Delta Q = m \times c \times \Delta T$ (given)	gas, constant V	$p_1/T_1 = p_2/T_2$
gas, constant T	$p_1V_1 = p_2V_2$	frequency	$f = 1 \div T$
Doppler/redshift	$\Delta\lambda/\lambda_0 = v/c$ (given)	orbital speed	$v = 2\pi r/T$ (given)

4 MOMENTUM

WORKED EXAMPLES

Jan 22: $p = 72 \times 13 = 936$ kg m/s; $F = 936 \div 0.29 = 3200$ N.

Jun 22 (explosion): $0 = 1.9 \times 3.7 + 2.9v \Rightarrow v = -2.4$ m/s (opposite direction).

Nov 24: $p = 39$ kg m/s $\Rightarrow v = 39 \div 8.1 = 4.8$ m/s; $F = 39 \div 0.56 = 70$ N.

Nov 25: $v = 120\,000 \div 14\,000 = 8.6$ m/s; $t = (120\,000 - 63\,000) \div 6100 = 9.3$ s.

Jan 20 (curling stones): p before $= 17 \times 2.9 \approx 49$; B's velocity from p after = p before; $F = \Delta p \div t$ with $t = 0.025$ s ≈ 1700 N.

MODEL ANSWER · CRUMPLE ZONES, AIRBAGS, CARPET (3) · JUN 22

The time of impact increases, so the rate of change of momentum decreases, so the force on the object (or person) is smaller.

5 MOMENTS

- Moment = force × perpendicular distance from the pivot: $28 \text{ N} \times 0.15 \text{ m} = 4.2 \text{ N m}$ (Jun 24); $3200 \times 4.8 = 15\,000 \text{ N m}$ (Jun 26).
- Principle of moments (Nov 20): $230 \times 0.37 = 0.98 \times F \Rightarrow F = 87$ N; (Jun 25) $1.14 = F \times 0.70 \Rightarrow F = 1.63$ N.
- Larger turning effect: larger force or longer handle (force further from the pivot).
- Supports (Nov 23, Jun 26): as a load moves towards support B, the force on B increases and on A decreases; the sum of the forces equals the total weight.
- Centre of gravity: the point where the whole weight appears to act.
- Work on a rotating wrench (Jun 24): distance = arc length $= 2\pi r \div 4$; work = force × distance.

6 ELECTROSTATICS AND ELECTROMAGNETISM

MODEL ANSWERS

Charging by friction (2): electrons are transferred by rubbing; the object that loses electrons becomes positive and the one that gains them becomes negative.

Earthing (3): electrons flow through the metal wire to (or from) earth until the object is neutral, preventing a spark, fire or explosion (refuelling aircraft, Jan 20).

Like charges repel, unlike attract; metal conducts, so charge spreads over it.

- $Q = It$: 6.8×10^{-6} A $\times 3.7 \times 10^{-3}$ s $= 2.5 \times 10^{-8}$ C (Nov 24). $E = QV$: $V = 3.7 \div 0.0043 = 860$ V (Jun 21).
- Electron speed from KE (Jun 23): $v = \sqrt{(2KE/m)} = 5.3 \times 10^{7}$ m/s.
- Electromagnet (Jan 22, Jun 26): a coil (solenoid) with an iron core and a current; stronger with more turns, more current, iron core.
- Transformer turns ratio (Jun 19): $40/N_s = 6.8/9.9 \Rightarrow N_s = 58$. Power: $275\,000 \times I_p = 230 \times 95 \Rightarrow I_p = 0.079$ A (Jan 21).

MODEL ANSWERS · TRANSFORMER (4) AND TRANSMISSION (5) · JUN 22, JAN 21

Alternating current in the primary coil produces a changing magnetic field; the iron core carries the field to the secondary coil; the changing field induces an alternating voltage in the secondary. Fewer secondary turns = step-down.

Step-up transformer before transmission: voltage up, current down, so less heating and energy loss in the cables; step-down transformer near homes for safety.

7 WAVES, SOUND AND THE OSCILLOSCOPE

FREQUENCY FROM AN OSCILLOSCOPE (5) · JAN 20, JUN 22, NOV 25

Connect a microphone to the oscilloscope; adjust to a steady trace with at least one complete cycle; count the squares for one (or several) cycles; multiply by the time base to get the period T; f = 1/T.

Jun 19: period = 4 squares × 0.25 ms = 1.0×10^{-3} s ⇒ f = 1000 Hz. A quieter, lower note has smaller amplitude and longer period.

SPEED OF SOUND METHODS (5–6) · JAN 19, JUN 19

Echo method: clap at least 50 m from a wall, time the echo (distance there and back); or see-and-hear method over 100 m; or two microphones 1 m apart on an oscilloscope. Use a tape measure and stopwatch, repeat and average, speed = distance ÷ time.

Jun 21: (300 − 120) ÷ 0.55 = 330 m/s; human reaction time is the main source of error.

- Wavelength: λ = 330 ÷ 25 000 = 0.013 m (Jun 21). Ultrasound is above 20 kHz.
- Doppler (Jun 19): a moving source gives shorter wavelengths ahead (higher frequency) and longer behind.
- Radar echo (Nov 20): distance = speed × time ÷ 2 = 285 km.

8 ENERGY RESOURCES AND TRANSFERS

Resource	Advantages	Disadvantages
solar	renewable, no CO_2, low maintenance	not at night, needs space, weather-dependent
hydroelectric	renewable, no fuel cost, quick response	flooding, reservoir level varies
nuclear	no CO_2, reliable, high energy density	radioactive waste, accident risk, decommissioning cost
fossil fuels	reliable, responds to demand	CO_2 (global warming), SO_2 (acid rain), non-renewable

- Hydroelectric chain (Jun 24): GPE store of water → kinetic store (mechanically) → turbine → generator → electrically to the grid.
- Nuclear chain (Jun 26): nuclear store of fuel → thermal store of coolant → kinetic store of turbine → generator.
- Power from energy per day: 9.7×10^{14} J ÷ 86 400 s = 1.1×10^{10} W (Nov 21).
- Solar charger (Jun 25): nuclear store of Sun → radiation → panel → electrically → chemical store of battery.

9 THERMAL PHYSICS AND GASES

SPECIFIC HEAT CAPACITY EXPERIMENT (6) · NOV 23, NOV 24, JUN 26

Measure the mass of water/block on a balance; measure the starting temperature; heat with an immersion heater for a measured time, recording V and I (energy = V × I × t, or a joulemeter); stir; record the maximum temperature after switching off; c = energy ÷ (m × ΔT); repeat and average; insulate and use a lid.

Nov 23: c = 54 000 ÷ (0.56 × 22) = 4400 J/kg °C. **Jun 26:** E = 12 × 3.9 × 80 = 3744 J; c = 3744 ÷ (0.29 × 28) = 460 J/kg °C.

Errors: heat lost to the surroundings, beaker and thermometer; gaps in the lid; so c comes out too high.

- Thermal energy: ΔQ = 0.016 × 4200 × 18 = 1210 J (Jan 23).
- Solids: regular arrangement, particles vibrate about fixed positions; liquids: irregular, close, slide past each other; gases: far apart, fast and random.

- Melting: energy weakens bonds between particles; temperature stays constant (plateau) until all has melted.
- Gas pressure: particles hit the walls; higher temperature ⇒ faster particles ⇒ more frequent and harder collisions.
- Pressure–temperature (Nov 21): $p_2 = 136 \times 230 \div 400 = 78$ kPa (kelvin throughout). Absolute zero: −273 °C, particles have minimum KE.
- Insulation (Nov 20): trapped air is a poor conductor and cannot circulate (no convection); white surfaces reflect radiation.

10 NUCLEAR PHYSICS

- Fission chain: a neutron is absorbed by U-235, which splits into daughter nuclei and 2–3 neutrons; these cause further fission.
- Fusion conditions (Nov 23): high temperature so nuclei have high kinetic energy to overcome repulsion (nuclei are positive, like charges repel); high pressure so nuclei are close together and collide often.
- Fusion vs fission: fusion joins light nuclei; fission splits heavy nuclei. A fusion product with no neutrons cannot sustain a chain (Nov 21).
- Alpha scattering (Jun 19): alpha is repelled, so the nucleus is positive like the alpha particle; F = ma gives $a = 5.5 \times 10^{26}$ m/s^2.

11 ASTROPHYSICS AND COSMOLOGY

HR DIAGRAM AND MAGNITUDES

Hertzsprung–Russell diagram: absolute magnitude (or luminosity) against colour or temperature (hot/blue on the left). Main sequence diagonal; red giants top right; white dwarfs bottom left.

Absolute magnitude: brightness of a star at a standard distance of 10 parsecs (32.6 light years). Two stars of the same absolute magnitude can appear different because they are at different distances.

BIG BANG (4–6) · JUN 19, JUN 22, JUN 25, NOV 25

Evidence: cosmic microwave background radiation (CMBR) and the red-shift of galaxies.

Red-shift: light from distant galaxies has a longer wavelength, so they are moving away; more distant galaxies show greater red-shift, so they move faster; the universe is expanding from a single point.

CMBR is the same in all directions; its wavelength has increased as the universe expanded and cooled.

Speed from redshift (Jun 23): $v = 3.0 \times 10^8 \times 15/630 = 7.1 \times 10^6$ m/s.

12 PRACTICAL METHODS AND FINAL CHECKLIST

- Bouncing ball vs temperature (Jan 23): water bath and thermometer, range of temperatures, measure bounce height at eye level, same drop height and surface, repeat and average.
- Black vs silver bottles (Jun 24): equal volumes, same distance from the heater, same starting temperature, measure after a fixed time, repeat; black absorbs radiation faster.
- Magnetic field and distance (Jan 22): ruler, change distance, measure field, keep current and coil constant, repeat.
- Density (Jun 25): balance (zeroed) for mass, displacement in a measuring cylinder for volume, $\rho = m/V$ with a unit.

1. Write the equation, convert to SI, substitute, rearrange, evaluate, unit.
2. Momentum: total before = total after; direction signs; $F = \Delta p/t$.
3. Moments: clockwise = anticlockwise about one point; perpendicular distance.

4. Charge: electrons move, never positive charges; earthing neutralises.
5. Gas laws in kelvin; heating curves have a plateau at melting.
6. Big Bang: CMBR + red-shift; absolute magnitude at 10 parsecs.
7. Methods: balance, timer, thermometer; control variables; repeats and mean; equation used.
8. Leave 5 minutes to check units and powers of ten.